Morning SPY hedge assessment

Report date: 2026-10-30

Verdict

The 2025 profit did not repeat descriptively in 2023. Across 196 expiry pairs, the main morning rule lost $48.96 per package after modeled costs.

Mean gross profit was $58.43 per package and modeled costs were $107.39. The 95% expiry-week cluster interval for net mean profit was -$221.01 to $117.67, so a positive expected profit is not established.

The morning rule improved the 2023 mean relative to the unhedged control by $75.58, but its paired confidence interval also includes zero. Residual overnight stock positions remained in 69 of 196 pairs.

This separate quote period does not establish a reliable after-cost edge. It is a historical daily-price calculation, with prior data exposure and execution limitations.

USD per package of ten short SPY option contracts and one long SPXW option contract. Figures are dollar P&L, not rates or annualized returns. Primary comparison uses full-cross, base costs, gross prefunding, and 1 bp stock slippage.

Main comparison

Period	n	Mean	Median	Total	95% CI	Win rate
2023 calendar year	196	-$48.96	-$66.70	-$9,596.60	[-$221.01, $117.67]	40.3%
2025 Jan-Aug	127	$653.39	-$10.08	$82,980.62	[$125.64, $1,274.85]	48.0%

2023 source coverage

Coverage check	Count
Expected pairs	196
Complete pairs	196
Pair/fill rows	588
Scenario cases	35280
Actual returned quote rows	68903
Projection fields matched	29

The statistical denominator is 196 pairs. The 588 pair/fill rows represent three hypothetical fill variants per pair, not 588 independent quote samples. The 35,280 scenario cases apply scenario combinations to those rows.

The 2023 replication is an additional historical sample, not a globally untouched or predictive holdout. The 2025 comparison covers January through August only, not a full year, and is not a paired inference against 2023.

2023 primary arms

All five arms use the same 196-pair sample. Absolute intervals measure each arm; paired intervals compare each arm with q0 on the shared sample.

Arm	n	Mean	Median	Absolute 95% CI	Paired vs q0 95% CI	Win rate
q0	196	-$124.54	-$17.15	[-$320.93, $75.75]	[$0.00, $0.00]	47.4%
fixed 500	196	-$89.95	-$150.68	[-$222.96, $45.40]	[-$124.66, $178.70]	44.4%
fixed 1,000	196	-$55.35	-$137.01	[-$267.47, $144.87]	[-$249.33, $357.40]	33.2%
morning-price rule	196	-$48.96	-$66.70	[-$221.01, $117.67]	[-$119.83, $273.89]	40.3%
assignment oracle	196	-$91.44	-$77.16	[-$114.53, -$70.07]	[-$164.47, $227.95]	32.7%

The oracle uses future assignment information. It is diagnostic only, not a tradable rule or profit bound.

Legacy period means

The historical comparison preserves the reviewed period means. The 2023 sample is shown separately and is not paired with 2025.

Period	n	Mean	Median	Total	95% CI	Win rate
2019 pilot	9	-$610.75	n/a	n/a	n/a	n/a
2022 H2	63	-$136.85	n/a	n/a	n/a	n/a
2024	195	-$90.03	n/a	n/a	n/a	n/a
2025 Jan-Aug	127	$653.39	-$10.08	$82,980.62	[$125.64, $1,274.85]	48.0%
2023 calendar year	196	-$48.96	-$66.70	-$9,596.60	[-$221.01, $117.67]	40.3%

Risk, contribution, and costs

Residual position counts, overnight price contribution, costs, notional, and drawdown describe the primary morning-price scenario.

Long positions	Short positions	Nonzero positions
33	36	69

Residual share exposure

Measure	Mean shares	Total shares
Signed residual shares	-15.31	-3,000.00

P&L contribution, mean and total USD

Measure	Mean USD	Total USD
Overnight price contribution	$82.96	$16,260.00
Gross P&L	$58.43	$11,452.43
Net P&L	-$48.96	-$9,596.60

Cost components, mean and total

Component	Mean	Total
Hedge funding	$48.03	$9,414.52
Option contract fees	$7.15	$1,401.40
Premium debit funding	$0.01	$1.96
Residual short borrow	$10.72	$2,102.06
Stock transaction fees	$4.34	$850.00
Stock slippage	$37.14	$7,279.09

Notional and drawdown

Maximum notional	Maximum peak-to-trough drawdown
$476,880.00	$23,950.32

Month cohorts

Month	n	Mean	Median	Total	Long positions	Short positions	Nonzero positions
2023-01	15	-$317.49	-$104.88	-$4,762.35	1	4	5
2023-02	15	$207.01	-$7.15	$3,105.16	2	4	6
2023-03	19	-$291.41	-$120.33	-$5,536.82	2	3	5
2023-04	15	$660.30	$76.66	$9,904.54	4	3	7
2023-05	17	-$165.66	-$53.58	-$2,816.28	3	1	4
2023-06	17	-$424.32	-$86.77	-$7,213.46	2	6	8
2023-07	14	-$131.94	-$111.86	-$1,847.10	2	3	5
2023-08	19	$95.49	-$17.15	$1,814.38	7	2	9
2023-09	16	-$856.56	-$145.14	-$13,705.02	3	2	5
2023-10	17	$542.12	$62.85	$9,215.96	4	2	6
2023-11	16	$11.85	-$104.38	$189.59	2	2	4
2023-12	16	$128.42	-$147.43	$2,054.79	1	4	5

Morning rule sensitivity

The table varies fill convention, base or stress costs, stock slippage, and funding treatment for the fixed morning-price rule.

Funding labels: Gross pre-funding means gross_prefunding. Assignment cash net means assignment_cash_net.

Fill	Cost	bp	Funding	Mean net USD	Win rate
Midpoint	Base	0	Gross pre-funding	$5.83	48.0%
Midpoint	Base	0	Assignment cash net	$38.05	56.1%
Midpoint	Base	1	Gross pre-funding	-$31.31	43.9%
Midpoint	Base	1	Assignment cash net	$0.91	52.0%
Midpoint	Base	5	Gross pre-funding	-$179.86	34.7%
Midpoint	Base	5	Assignment cash net	-$147.64	37.8%
Midpoint	Stress	0	Gross pre-funding	-$80.92	38.8%
Midpoint	Stress	0	Assignment cash net	-$16.49	52.6%
Midpoint	Stress	1	Gross pre-funding	-$118.06	35.7%
Midpoint	Stress	1	Assignment cash net	-$53.63	46.9%
Midpoint	Stress	5	Gross pre-funding	-$266.61	31.1%
Midpoint	Stress	5	Assignment cash net	-$202.18	33.2%
Halfway to cross	Base	0	Gross pre-funding	-$3.00	45.9%
Halfway to cross	Base	0	Assignment cash net	$29.22	54.6%
Halfway to cross	Base	1	Gross pre-funding	-$40.13	42.3%
Halfway to cross	Base	1	Assignment cash net	-$7.92	50.0%
Halfway to cross	Base	5	Gross pre-funding	-$188.69	33.7%
Halfway to cross	Base	5	Assignment cash net	-$156.47	36.7%
Halfway to cross	Stress	0	Gross pre-funding	-$89.75	38.3%
Halfway to cross	Stress	0	Assignment cash net	-$25.32	52.0%
Halfway to cross	Stress	1	Gross pre-funding	-$126.89	35.2%
Halfway to cross	Stress	1	Assignment cash net	-$62.46	44.9%
Halfway to cross	Stress	5	Gross pre-funding	-$275.44	30.6%
Halfway to cross	Stress	5	Assignment cash net	-$211.01	32.7%
Full cross	Base	0	Gross pre-funding	-$11.82	43.9%
Full cross	Base	0	Assignment cash net	$20.39	52.6%
Full cross	Base	1	Gross pre-funding	-$48.96	40.3%
Full cross	Base	1	Assignment cash net	-$16.75	48.0%
Full cross	Base	5	Gross pre-funding	-$197.52	32.1%
Full cross	Base	5	Assignment cash net	-$165.30	34.2%
Full cross	Stress	0	Gross pre-funding	-$98.58	37.2%
Full cross	Stress	0	Assignment cash net	-$34.15	49.5%
Full cross	Stress	1	Gross pre-funding	-$135.72	34.2%
Full cross	Stress	1	Assignment cash net	-$71.28	43.4%
Full cross	Stress	5	Gross pre-funding	-$284.27	29.6%
Full cross	Stress	5	Assignment cash net	-$219.84	31.6%

Method and limitations

The fixed rule sets q to 1,000 shares when the official SPY open on the expiry session is at least the frozen entry-selected SPY strike plus $0.01; otherwise q is zero. The comparison tolerance is 1e-10. The morning price input chooses hedge size. It is not the modeled purchase price. The modeled hedge is bought at expiry close, then residual stock is closed at the next open. The 2023 quote clock is unverified. Some 2023 option examples and 2019–2025 stock history were inspected earlier; the entire new 68,903-row quote export was not. The 2023 sample is an additional historical replication, not a globally untouched or predictive holdout. The 2025 sample covers January through August only, not a full year, and is not paired inference against 2023. Daily theoretical fills do not prove execution, capacity, funding availability, or margin. Physical assignment is a proxy; early assignment remains possible. Do not extend the 2024/2025 15:59 quote claim to 2023. Cboe actual SPXW PM settlement is not ordinary SPX close. No tuned rule, profit guarantee, or trading recommendation is implied.

Costs: 11 option contracts at $0.65 base / $1.50 stress each; stock fees $0.005 / $0.01 per share; annual borrow 3% / 10%; funding 5% / 10%; SPY and SPXW option contract multipliers are 100; package ratio is ten SPY options short to one SPXW option long. Bootstrap uses shared expiry ISO week clusters, 10,000 draws, seed 20261030, linear type-7 quantiles.

Sources: WRDS queries 11847843 and 11847851; Cboe W/M/S 2023 settlement files https://www.cboe.com/index_settlement_values/get_sv_data/W/2023/, https://www.cboe.com/index_settlement_values/get_sv_data/M/2023/, https://www.cboe.com/index_settlement_values/get_sv_data/S/2023/.

Provenance

Validation PASS, 1777327 checks, receipt SHA-256 `4e297cb6a8c941c9dce33532c7355aad4d3ce0d39d30a105d42aeab1e4b0cba3`; max error 5.2523319027386606e-11.

Git 64e72887b782fdf838257a09877ffb817dfbfbdf clean=True; displayed command (private paths redacted): [local path] -m spyspx.morning_hedge_2023 evaluate --input-manifest [local path] --input-manifest-sha256 `47972793ea9daadfa35e5fa7aea884f61759d750525384946f18994dc1b89656` --selection [local path] --selection-sha256 `00cff4fb8c3ec13d9ea72cc4ee4d4c5cc1723747159bdea723305e2c6a167f13` --out planning/morning-hedge-replication-20261030/evaluation-01. The exact executed command is retained in the bound private execution receipt.

Input manifest `47972793ea9daadfa35e5fa7aea884f61759d750525384946f18994dc1b89656`; protocol `8f66787ce63d47771c1bbb5edd516c02d46ebc37b2538f3581ae3e073cbfa403`; calendar `122296e11825bbdec9af5bbed5cca5f3fc6c8ea0dc99548351dc05d44beee809`; selection `00cff4fb8c3ec13d9ea72cc4ee4d4c5cc1723747159bdea723305e2c6a167f13`; adapter output `6cf57c05e5ab8151d2e711fbff39437a93b821320def08141565460a15f3bb8c`; execution `3b50a9eea10ee0574f9fb2f6162ff491463503c50107415448d34a9865cc766a`.

Adapter code hashes: code_adapter=`02cd50de29e5c843a2bbace14ce1825d6027ea2268eec105878396a2030b8a13`, code_admission=`c9ab3d063535b588ab0f9d42f2d8549e6f6d711942a6aff2cc2cca9f8439a506`, code_expiry_extension=`6a5964a78334cbe2d8fa3f1285221223eb609daeb39061036e62a022f9c481d0`, code_expiry_payoff=`c7eaf0c90fd5e029dac3012630af88646c0c8329697279a198ffd4842a8fbb48`, code_normalizer=`045971a5b41c83d311b40f6733a7c20578b5e326ada4fdbc0417a9e1506a8130`, code_overnight_hedge=`6cca6772325e7d55c8a85ef6a444681d2a54175e9ddfb10949dfdae6748337f3`, code_replication=`c010d765e58d0130b3f61b4253a8e1e08ca068b5e7a920c69e424e12952e59b8`.

Legacy statistics `295078e0fa11b339cfa995786b9536ca8240c8ff98c9911f33684ec69ccfcb55`; render input `fd15e0ec4f75c9d854a345533f3565fd429b8f443ce2ebb702495764c21f837d`; renderer `1e14792462300df2f204564846b6af1c5167cbc067094a0391c94ea3f80248f0`; PDF recorded in Markdown and render receipt.